



Semantic Web

Lab1 – XML Revision

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Definition

What is XML?

- eXtensible Markup Language
- Containing structured / semi-structure information
- Defined by four specifications:

- ☐ XML, the Extensible Markup Language
- ☐ XLL, the Extensible Linking Language
- ☐ XSL, the Extensible Style Language
- ☐ XUA, the XML User Agent

(Help to standardize software supporting XML, e.g. browsers.)



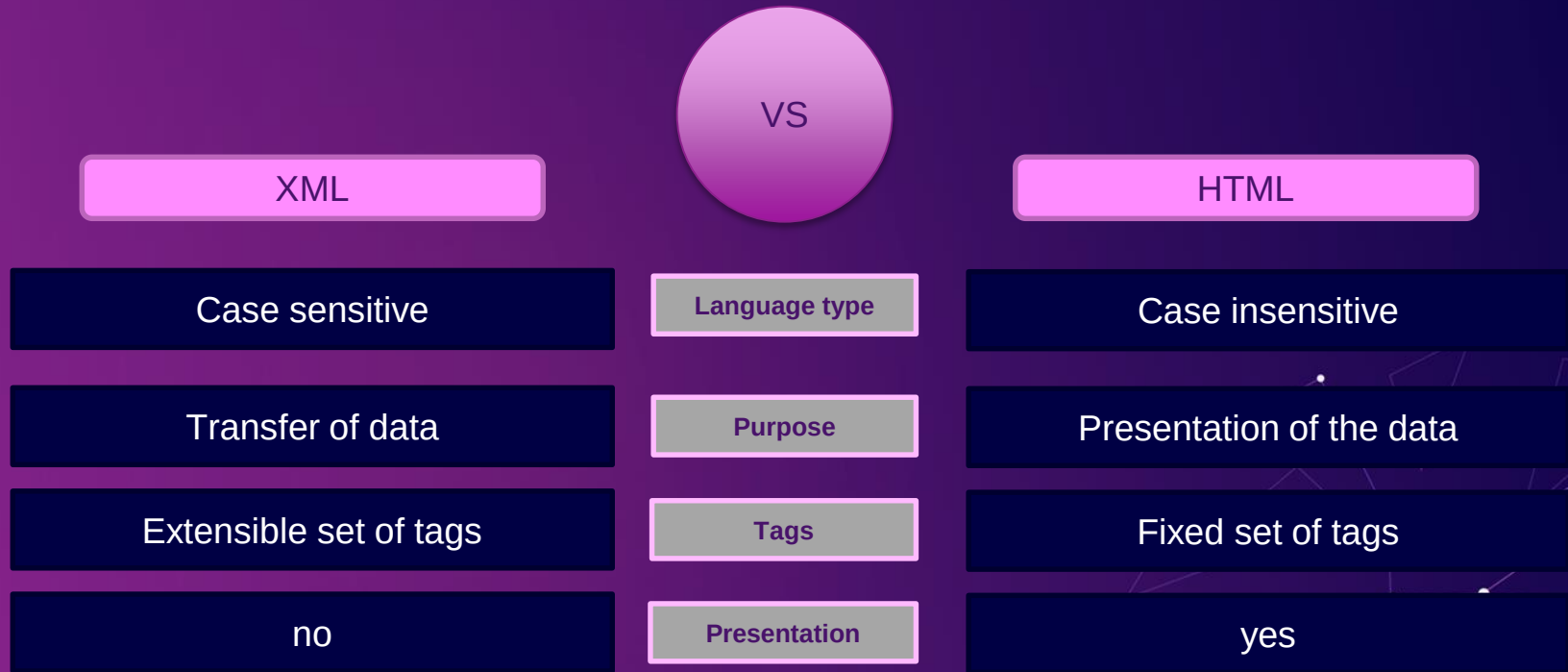
XML...

XML is **Not**:

- A programming language
- A presentation format
- A replacement for HTML (but HTML can be generated from XML)
- A network transfer protocol (but XML may be transferred on network)
- A database (but XML may be stored into a database)



COMPARISON BETWEEN XML VS. HTML



XML SYNTAX

An XML element is made up of a start tag, an end tag, and data in between.

```
<article>  
<author>Gerhard Weikum</author>  
<title>The Web in 10 Years</title>  
</article>
```

- Easy to understand for human users
- Well structured, easy to read and write from programs

XML tags are case-sensitive:

1

`<CITY> != <City> != <city>`

2

Some special characters must be escaped using entities:

&	→	&
<	→	<
>	→	>
"	→	"
'	→	&apos

XML NAMESPACE

Solving the Name Conflict Using a Prefix

```
<table>  
<name>African Coffee Table</name>  
<width>80</width>  
<length>120</length>  
</table>
```

Can be?



XML NAMESPACE

Solving the Name Conflict Using a Prefix



```
<root>
```

```
  <h:table id="500">
```

```
    <h:name>Apples</h:name>
```

```
    <h:type>Bananas</h:type>
```

```
  </h:table>
```

```
  <f:table id="200">
```

```
    <f:name>African Coffee Table</f:name>
```

```
    <f:width>80</f:width>
```

```
    <f:length>120</f:lengt>
```

```
  </f:table>
```

```
</root>
```

Can be?



XML Example

```
<BOOKS>
```

```
  <book id="123" loc="library">
```

```
    <author>jone</author>
```

```
    <title>California</title>
```

```
    <year> 1995 </year>
```

```
  </book>
```

```
  <article id="555" ref="123" >
```

```
    <author>mo</author>
```

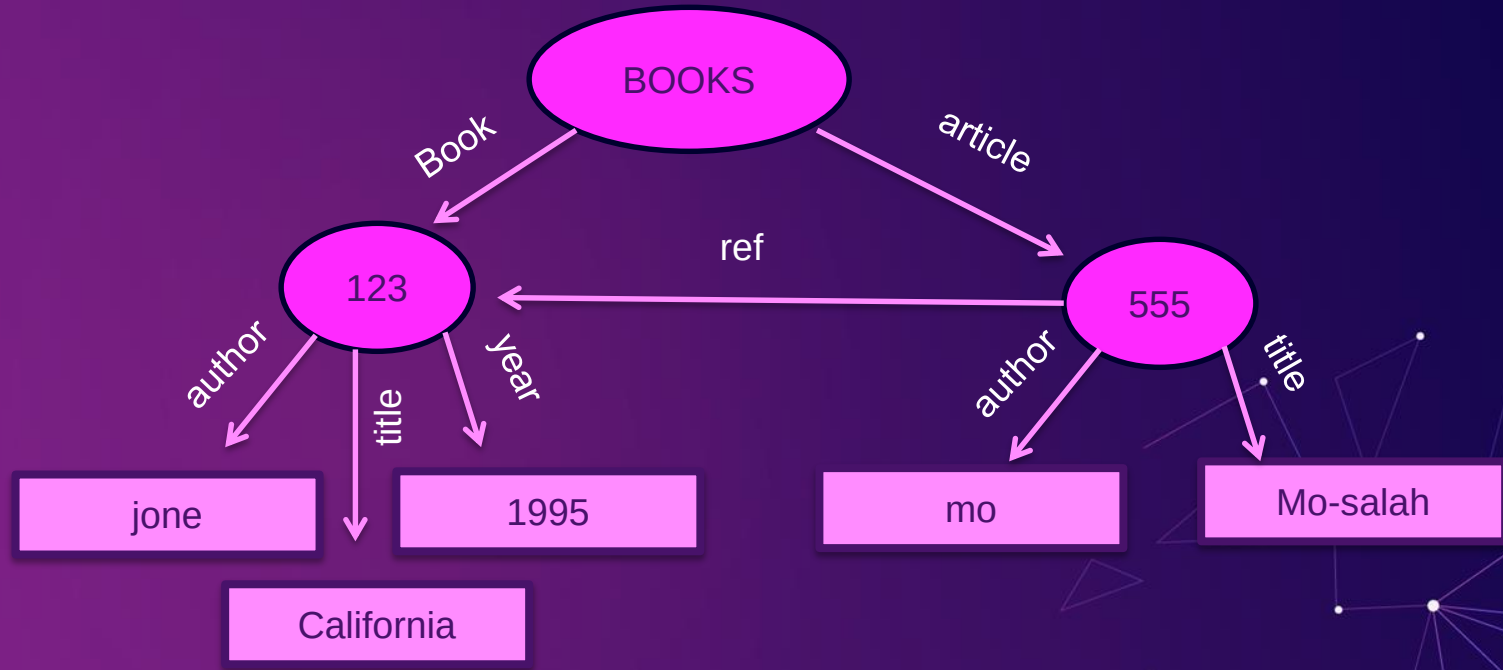
```
    <title> mo-salah</title>
```

```
  </article>
```

```
</BOOKS>
```



XML Example



XML Document object Model (DOM)

- Represents the XML document as a document tree.
- It gives access to the XML document (Nodes and Elements, and it has its own properties).
- Make XML structure to be editable (Allow adding, changing and removing values)



DOCUMENT TYPE DEFINITION (DTD)

A Document Type Definition (DTD) defines the structure and the validation of elements and attributes of an XML document

DTD has the form:



```
<!DOCTYPE Root_name [  
<!ELEMENT Root_name (child_name)>  
<!ELEMENT child_name (#DATA)> ]>
```



DTD Example

```
<article> <title> DTD </title> </article>
```

```
< ! DOCTYPE article[  
< ! ELEMENT article (title)>  
< ! ELEMENT title(#DATA)> ]>
```



DTD Example

```
<db>
  <person>
    <name> Alan</name>
    <age>42</age>
    <email> agb@usa.net </email>
  </person>
  <person>
    .....
  </person>
</db>
```

Solution :

```
< ! DOCTYPE db [
< ! ELEMENT db (person*)>
< ! ELEMENT person(name,age,email)>
< ! ELEMENT name(Alan)>
< ! ELEMENT age(42)>
< ! ELEMENT email(agb@usa.net)>
]>
```



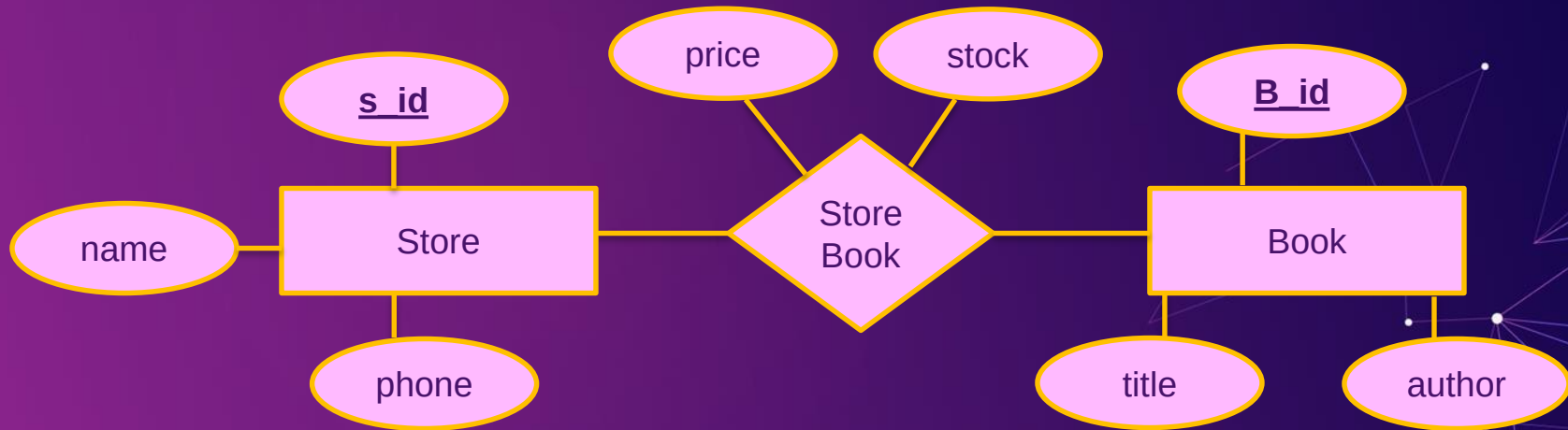
RELATIONAL DATABASE TO XML

Relational Database:

Store (s-id, name, phone)

Book (b-id, title, authors)

StoreBook (s-id , b-id, price, stock)





THANKS

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